

AP CALCULUS AB – UNIT 3

Differentiation: Composite, Implicit, and Inverse Functions

Expanded AP-Style Review: Full Coverage and Challenge



Name: _____ Class: _____ Date: _____

Directions

This review deliberately samples every website lesson and includes a larger proportion of difficult and challenge reasoning. Part A is non-calculator. Part B is calculator-permitted. Free-response work must show setup, exact reasoning, units, and theorem conditions.

Section I – Part A: No Calculator

1. If $F(x) = \sin((1 + x^2)^4)$, then $F'(x)$ is

(A) $4(1 + x^2)^3 \cos x$

(C) $\cos(4(1 + x^2)^3)$

(B) $8x(1 + x^2)^3 \sin((1 + x^2)^4)$

(D) $8x(1 + x^2)^3 \cos((1 + x^2)^4)$

2. For $x^2 + xy + y^2 = 7$, dy/dx is

(A) $-\frac{2x + y}{x + 2y}$

(C) $-\frac{2x}{2y}$

(B) $-\frac{x + 2y}{2x + y}$

(D) $\frac{2x + y}{x + 2y}$

3. If $f(3) = 8$ and $f'(3) = -4$, then $(f^{-1})'(8)$ is

(A) $\frac{1}{8}$

(C) -4

(B) $-\frac{1}{4}$

(D) $\frac{1}{4}$

4. Given $g(1) = 2$, $g'(1) = -3$, and $f'(2) = 5$, $(f \circ g)'(1)$ equals

(A) 8

(C) 10

(B) -15

(D) 2

5. An isolated formula for y' gives $0/0$ at a point on an implicit curve. The best conclusion is

(A) The tangent is automatically vertical

(C) The derivative formula is inconclusive and the original relation must be analyzed

(B) The point cannot lie on the curve

(D) The tangent is automatically horizontal

6. $\frac{d}{dx} \arcsin(2x)$ equals

(A) $\frac{1}{\sqrt{1 - 4x^2}}$

(C) $\frac{1}{2\sqrt{1 - x^2}}$

(B) $\frac{2}{1 + 4x^2}$

(D) $\frac{2}{\sqrt{1 - 4x^2}}$

7. Differentiate $x(x^2 + 3)^2$.

(A) $2x(x^2 + 3)$

(C) $4x(x^2 + 3)$

(B) $(x^2 + 3)^2$

(D) $(x^2 + 3)(5x^2 + 3)$

8. For the implicit curve $x^2 + xy + y^2 = K$, which formula for dy/dx is correct?

(A) $-\frac{x+2y}{2x+y}$

(C) $\frac{2x+y}{x+2y}$

(B) $-\frac{2x+1}{1+2y}$

(D) $-\frac{2x+y}{x+2y}$

9. If $f(4) = 3$ and $f'(4) = 2$, find $(f^{-1})'(3)$.

(A) 2

(C) $\frac{1}{3}$

(B) $-\frac{1}{2}$

(D) $\frac{1}{2}$

10. Differentiate $x(x^2 + 3)^4$.

(A) $(x^2 + 3)^4$

(C) $(x^2 + 3)^3(9x^2 + 3)$

(B) $8x(x^2 + 3)^3$

(D) $4x(x^2 + 3)^3$

11. For the implicit curve $x^2 + xy + y^2 = K$, which formula for dy/dx is correct?

(A) $-\frac{2x+y}{x+2y}$

(C) $\frac{2x+y}{x+2y}$

(B) $-\frac{x+2y}{2x+y}$

(D) $-\frac{2x+1}{1+2y}$

12. Differentiate $\arcsin(3x)$.

(A) $\frac{3}{1+9x^2}$

(C) $\frac{1}{\sqrt{1-9x^2}}$

(B) $\frac{3}{\sqrt{1-9x^2}}$

(D) $-\frac{3}{\sqrt{1-9x^2}}$

13. Find $\frac{d}{dx}[\sin(3x^2)]$.

(A) $6x \sin(3x^2)$

(C) $3 \cos(3x^2)$

(B) $\cos(3x^2)$

(D) $6x \cos(3x^2)$

14. For the implicit curve $x^2 + xy + y^2 = K$, which formula for dy/dx is correct?

(A) $\frac{2x+y}{x+2y}$

(C) $-\frac{2x+1}{1+2y}$

(B) $-\frac{x+2y}{2x+y}$

(D) $-\frac{2x+y}{x+2y}$

15. Differentiate $\arctan(x^2)$.

(A) $\frac{1}{1+x^4}$

(C) $\frac{2x^1}{1+x^2}$

(B) $\frac{2x^1}{\sqrt{1-x^4}}$

(D) $\frac{2x}{x^4+1}$

16. Differentiate e^{4x^3} .

(A) $12x^2e^{4x}$

(C) e^{4x^3}

(B) $12x^2$

(D) $12x^2e^{4x^3}$

17. For $x^2 + y^2 = 13$, find the slope at $(2, 3)$.

(A) $-\frac{3}{2}$

(C) $\frac{3}{2}$

(B) $-\frac{2}{3}$

(D) $\frac{2}{3}$

18. If $f(3) = 1$ and $f'(3) = 2$, find $(f^{-1})'(1)$.

- (A) 2
(B) $\frac{1}{3}$
(C) $-\frac{1}{2}$
(D) $\frac{1}{2}$

19. Differentiate $\ln(x^2 + 1)$.

- (A) $2x \log(x^2 + 1)$
(B) $\frac{1}{x^2+1}$
(C) $\frac{2}{x^2+1}$
(D) $\frac{2x}{x^2+1}$

20. For $x^2 + y^2 = 25$, find the slope at $(4, 3)$.

- (A) $-\frac{3}{4}$
(B) $-\frac{4}{3}$
(C) $\frac{3}{4}$
(D) $\frac{4}{3}$

21. Differentiate $\arcsin(3x)$.

- (A) $-\frac{3}{\sqrt{1-9x^2}}$
(B) $\frac{3}{\sqrt{1-9x^2}}$
(C) $\frac{1}{\sqrt{1-9x^2}}$
(D) $\frac{3}{1+9x^2}$

22. Find $\frac{d}{dx}[\sin(3x^2)]$.

- (A) $3 \cos(3x^2)$
(B) $6x \cos(3x^2)$
(C) $6x \sin(3x^2)$
(D) $\cos(3x^2)$

23. For $x^2 + y^2 = 20$, find the slope at $(2, 4)$.

- (A) 2
(B) $\frac{1}{2}$
(C) -2
(D) $-\frac{1}{2}$

24. Differentiate $\arctan(x^2)$.

- (A) $\frac{1}{1+x^4}$
(B) $\frac{2x^1}{\sqrt{1-x^4}}$
(C) $\frac{2x}{x^4+1}$
(D) $\frac{2x^1}{1+x^2}$

Section I – Part B: Calculator Permitted

- 25.** Use a calculator to evaluate the derivative of $\sin(4x)$ at $x = -1.0$.
- (A) -2.615 (B) 3.027
(C) 2.615 (D) -0.654
- 26.** A calculator identifies the point $(16/3, 8/3)$ on $x^3 + y^3 = 12xy$. The tangent slope there is closest to
- (A) 0.80 (B) -1.25
(C) -0.80 (D) 1.25
- 27.** A table shows $f(2.4) = 7.1$ and $f'(2.4) = 0.38$. Then $(f^{-1})'(7.1)$ is closest to
- (A) 18.684 (B) 2.632
(C) 0.38 (D) -2.632
- 28.** Use a calculator to evaluate the derivative of $\sin(3x)$ at $x = 1.04$.
- (A) 0.065 (B) -1.0
(C) -2.999 (D) 2.999
- 29.** For $x^2 + xy + y^2 = K$, estimate dy/dx at $(1.17, 1.05)$.
- (A) -1.037 (B) 1.037
(C) -0.965 (D) -1.114
- 30.** Find $\frac{d}{dx}(\arcsin x)$ at $x = -0.11$ to three decimals.
- (A) -0.11 (B) 0.89
(C) -1.006 (D) 1.006
- 31.** Use a calculator to evaluate the derivative of $\sin(2x)$ at $x = 0.66$.
- (A) 1.937 (B) 0.248
(C) -0.496 (D) 0.496
- 32.** For $x^2 + xy + y^2 = K$, estimate dy/dx at $(0.65, 0.62)$.
- (A) 1.016 (B) -1.048
(C) -0.984 (D) -1.016
- 33.** A numerical table gives $f(3.6) = 6.9$ and $f'(3.6) = 1.3$. Estimate $(f^{-1})'(6.9)$.
- (A) 1.3 (B) 0.145
(C) -0.769 (D) 0.769
- 34.** Use a calculator to evaluate the derivative of $\sin(4x)$ at $x = 0.16$.
- (A) 2.389 (B) 0.802
(C) 3.208 (D) -3.208
- 35.** For $x^2 + xy + y^2 = K$, estimate dy/dx at $(1.86, 0.86)$.
- (A) 1.279 (B) -0.782
(C) -2.163 (D) -1.279
- 36.** Find $\frac{d}{dx}(\arcsin x)$ at $x = -0.01$ to three decimals.
- (A) -1.0 (B) 1.0
(C) -0.01 (D) 0.99

Section II – Free Response

Questions 1–2 may be completed with a calculator. Questions 3–4 are non-calculator.

FRQ1 – Composition and inverse data

	x	1	2	4	7
	$f(x)$	4	7	1	2
The table gives	$f'(x)$	−3	5	2	−1
	$g(x)$	2	4	7	1
	$g'(x)$	6	−2	3	4

. Assume f is one-to-one.

(a) Find $(f \circ g)'(1)$ and $(g \circ f)'(1)$. [4 points]

(b) Find $(f^{-1})'(7)$. [2 points]

(c) Find the tangent line to $y = f^{-1}(x)$ at $x = 7$. [2 points]

FRQ2 – Implicit curve and tangent classification

The curve $x^2 + xy + y^2 = 7$ is considered.

(a) Find dy/dx . [3 points]

(b) Find all points with horizontal tangents. [3 points]

(c) Find all points with vertical tangents. [3 points]

FRQ3 – Multi-layer differentiation

$$\text{Let } F(x) = \left[1 + \ln \left(1 + e^{x^2} \right) \right]^5.$$

(a) Identify the successive inner functions from the deepest layer outward. [2 points]

(b) Find $F'(x)$. [4 points]

(c) Explain why $F'(0) = 0$ without evaluating all other factors numerically. [2 points]

FRQ4 – Second implicit derivative

On the upper semicircle $x^2 + y^2 = 25$, consider the point $(3, 4)$.

(a) Find y' and evaluate it at the point. [2 points]

(b) Derive a formula for y'' in terms of y and y' . [3 points]

(c) Find $y''(3)$ and interpret the concavity locally. [2 points]